M2 Component: Refuse-Tea Prediction & System Integration

Exploratory Data Analysis (EDA) & Pre-Modeling Audit Report

Project: AI-Based Decision Support System for Sri Lankan Tea Export Operations | **Author:** Senior Data Science Team
Dataset: M2_refuse_tea_synthetic_dataset.csv (620 Batches, 19 Columns) | **Date:** October 2026 | **Status:** Pre-Modeling EDA Complete

CRITICAL NOTICE — SYNTHETIC DATASET DISCLOSURE:
The dataset analyzed in this report (620 batch-level records) is a **synthetic dataset** generated for development, system architecture validation, and machine learning pipeline testing prior to live factory telemetry collection. All quantitative findings, distributions, and empirical correlations reflect the generator's parameterized specifications and must **not** be cited as real empirical findings about Sri Lankan tea factories.

Executive Summary & Core Directives

This report provides the pre-modeling exploratory data analysis for the M2 refuse-tea prediction component. The goal is to accurately predict the percentage of refuse tea generated per batch based on leaf characteristics, environmental variables, and factory processing parameters. Key insights include:

- **Elevated Target Distribution:** The target variable (refuse_tea_pct) averages **6.59%** (median: 6.38%, std: 1.52%, range: 3.29% to 14.25%), running +0.59% points above the 6.0% national industry baseline. A substantial **12.9% of batches exceed the 8.0% adverse loss threshold**, and 3.4% experience severe loss (>10.0%), confirming a pronounced right tail.
- **Hidden Non-Linear V-Shaped Curves:** Conventional linear correlation indicates near-zero signal for withering_duration_hrs (r = -0.028) and dryer_temperature_c (r = +0.025). However, binned non-linear analysis proves they follow distinct **U-shaped / V-shaped curves** with clear optimal operational windows (withering: 16.5–17.5 hrs; drying: 109–113 °C). Non-linear models like XGBoost are essential.
- **Mandatory Target Leakage Exclusion:** made_tea_output_kg and refuse_tea_kg are post-hoc scale weights mathematically derived from the target. Including either creates catastrophic 100% data leakage. **Both must be excluded in preprocessing.py.**
- **Multicollinearity & Post-Hoc Variables:** region and elevation_band have a 100% 1-to-1 bijection (Low-Country = LOW, Uva = MEDIUM, Nuwara Eliya = HIGH); elevation_band must be dropped. refuse_resale_status is a post-production classification and must be excluded from feature inputs.
- **High Data Quality:** Clean dataset with 0 missing values, 0 duplicate batch IDs, and realistic mass conservation (mean outturn ratio of 22.33%).

1. Data Quality & Sanity Audit

The dataset contains 620 batch records spanning two full calendar years (2024–2025) across 8 tea factories. Standard data hygiene checks confirm full structural integrity without missing values or duplicate identifiers.

Metric / Audit Check	Observed Value	Assessment & Preprocessing Directive
Dataset Dimensions	620 rows × 19 columns	Sufficient sample size for 5-fold cross-validation.
Missing Value Audit	0 missing across all 19 columns	Complete dataset; zero imputation required.
Primary Key Integrity	0 duplicates in batch_id (B0001–B0620)	Unique index confirmed; drop batch_id before modeling.
Mass Range Validation	Green leaf: 1,950–16,495 kg; Made: 418.5–3,415.4 kg	Zero negative masses; physical consistency verified.
Outturn Ratio Sanity	Mean: 22.33% (Min: 19.00%, Max: 25.63%)	Precisely matches orthodox dry tea recovery norm (21%–23%).
Target Boundary Sanity	Range: [3.29%, 14.25%]	Fully within legitimate [0, 100] theoretical percentage bounds.

2. Target Variable Analysis: refuse_tea_pct

The target variable represents the percentage of total dry tea sorted out as unmarketable refuse (fibrous stalks, unfermented residue, fluff, and scorched particles). The sample distribution has a positive right skew (+1.14), meaning most batches achieve acceptable quality, while an adverse subset drives disproportionate yield losses.

Statistical Parameter	Observed Value	Industrial Benchmark / Interpretation
Mean / Std Dev	6.588% ± 1.525%	+0.588% pts higher than the 6.00% Sri Lanka Tea Board baseline.
Median / IQR	6.380% (IQR: 5.570% – 7.253%)	50% of batches cluster tightly within a 1.68% percentage point band.
Distribution Skewness	+1.140 (Kurtosis: +2.706)	Right-skewed tail; high-waste events penalize standard RMSE.
Batches > 6.0% (National Base)	413 batches (66.6%)	Two-thirds of processed batches exceed ideal benchmark efficiency.
Batches > 8.0% (Adverse Alert)	80 batches (12.9%)	Severe profit margin erosion zone requiring proactive operator alerts.
Batches > 10.0% (Critical Failure)	21 batches (3.4%)	Peak at 14.25%; mechanical failure or extreme weather combinations.

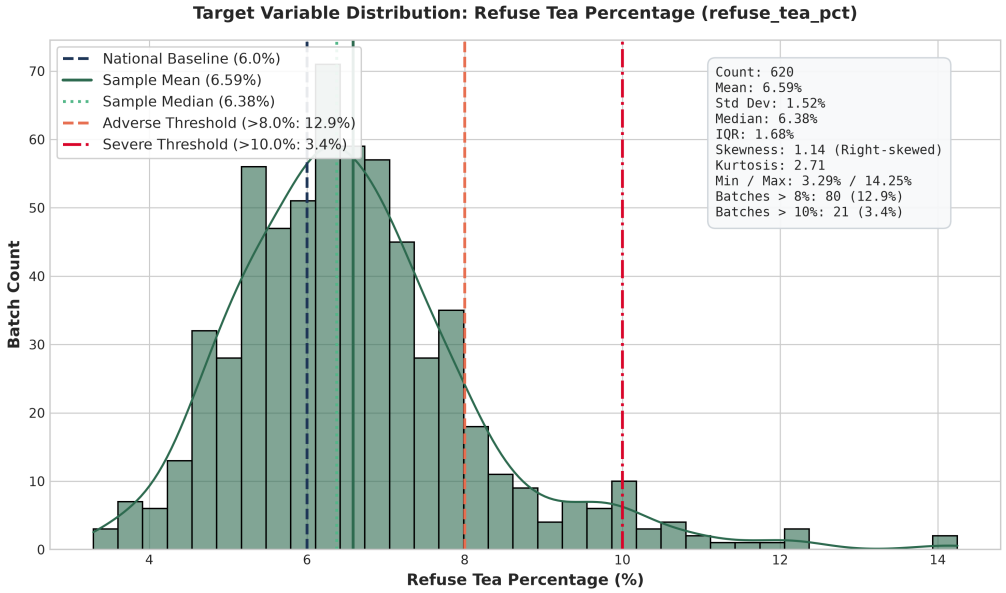


Figure 1: *Empirical distribution of refuse_tea_pct with national baseline (6.0%) and adverse thresholds (>8%, >10%).*

3. Univariate Feature Distributions

Operational and ambient variables exhibit distinct distributions reflecting factory physics and tea agro-ecology.

Feature Name	Mean ± Std	Min – Max	Skew	Outliers	Operational Impact
green_leaf_input_kg	7,182 ± 2,739	1,950 – 16,495	+0.52	2 (0.3%)	Trough loading pressure during flush days.
plucking_fine_leaf_pct	78.17 ± 6.81%	56.2 – 96.0%	-0.06	6 (1.0%)	Symmetric; coarser leaf introduces woody stalk.
withering_duration_hrs	16.89 ± 2.27 h	10.0 – 23.3 h	-0.04	10 (1.6%)	Centered at standard trough residence (16.9 h).
dryer_temperature_c	110.54 ± 6.91 °C	94.5 – 134.2 °C	+0.16	3 (0.5%)	Centered at standard firing temperature (110 °C).
machine_downtime_hrs	0.64 ± 0.61 h	0.0 – 4.39 h	+1.73	34 (5.5%)	Heavy right skew; breakdowns cause leaf stewing.
ambient_temperature_c	22.96 ± 4.48 °C	11.9 – 31.8 °C	-0.19	0 (0.0%)	Multi-modal spread reflecting elevation bands.
relative_humidity_pct	77.33 ± 10.17%	37.4 – 97.0%	-0.89	19 (3.1%)	Left-skewed; low values in dry Uva season.

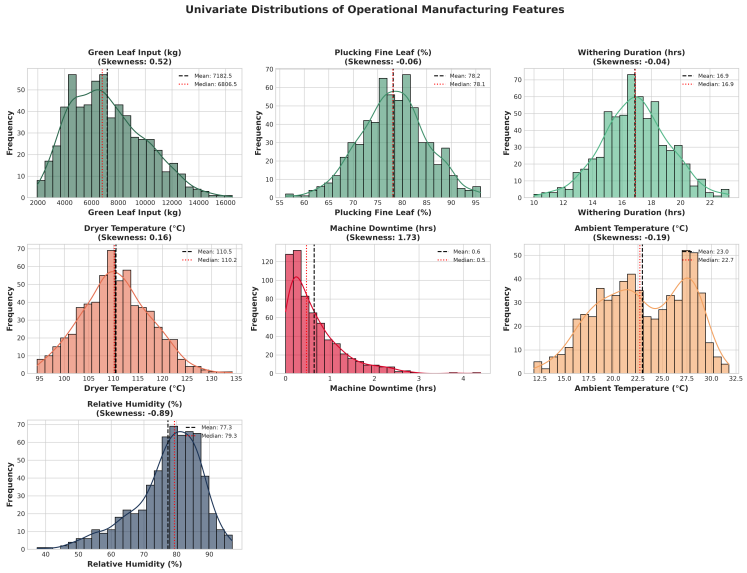


Figure 2: *Univariate histograms and kernel density estimates for operational manufacturing features.*

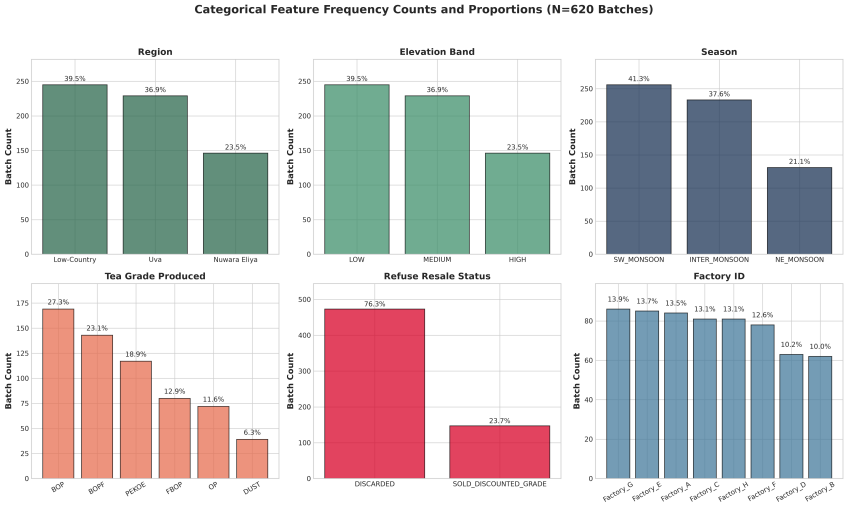


Figure 3: *Distribution of categorical dimensions (Region, Elevation Band, Season, Grade, Resale, Factory).*

4. Bivariate Relationships & Non-Linear Patterns

Linear Pearson correlation (r) and non-parametric Spearman rank correlation ($ ho$) reveal operational drivers. Critically, linear correlation fails to detect two of the most important process parameters due to V-shaped response curves.

Operational Feature	Pearson r	Spearman ρ	Curve Type	Underlying Industrial Mechanism
machine_downtime_hrs	+0.379	+0.335	Monotonic (+)	Stoppages delay processing; rolled leaf ferments and stews.
relative_humidity_pct	+0.246	+0.330	Monotonic (+)	High ambient moisture impedes withering, leaving turgid stalks.
plucking_fine_leaf_pct	-0.170	-0.188	Monotonic (-)	Fine leaf yields tender shoots; coarse plucking adds woody fiber.
dryer_temperature_c	**+0.025**	**+0.025**	**V-Shaped Curve**	**Optimum at 109–113 °C.** Under-drying causes stewing; over-drying scorches.
withering_duration_hrs	**-0.028**	**-0.036**	**U-Shaped Curve**	**Optimum at 16.5–17.5 h.** Under-wither tears leaf; over-wither desiccates.

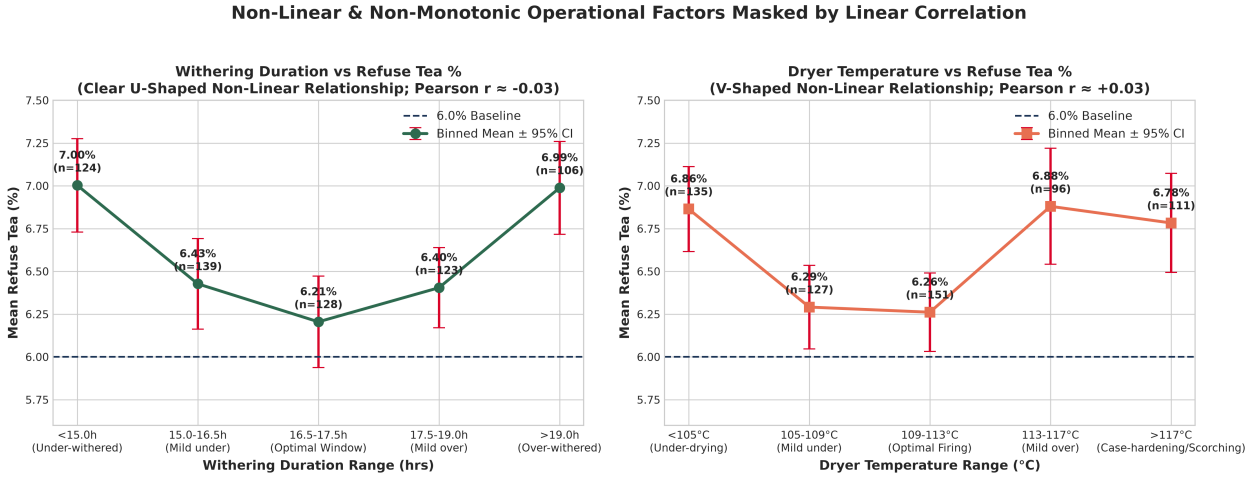


Figure 4: *Binned mean refuse percentage displaying the U-shaped withering response and V-shaped dryer response.*

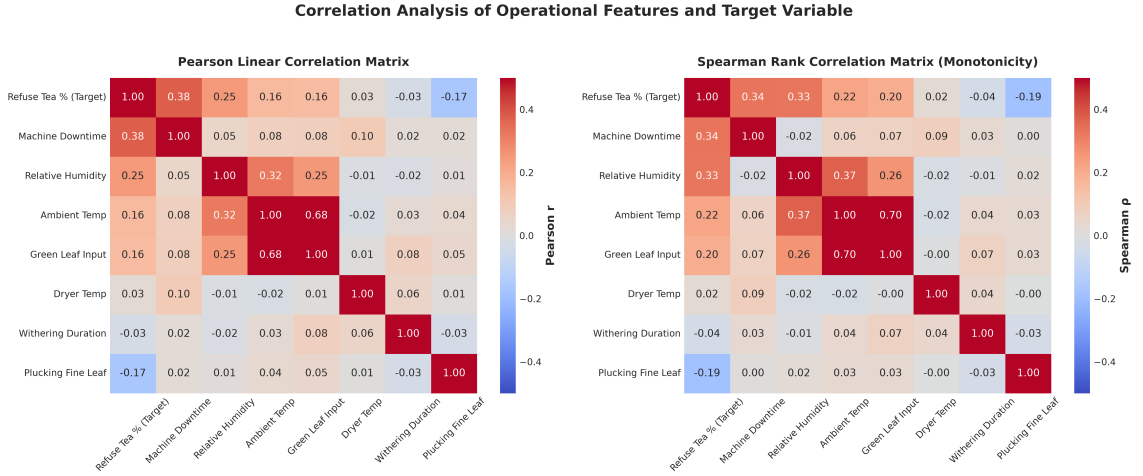


Figure 5: *Pearson linear vs Spearman rank correlation heatmaps across all operational numerical features.*

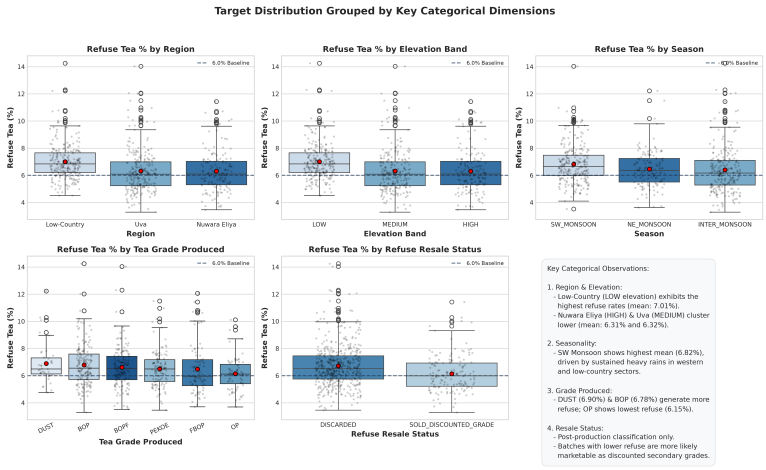


Figure 6: *Target distribution across Region, Elevation Band, Season, Grade Produced, and Resale Status.*

5. Grouped & Multi-Factor Dynamics

Multi-factor breakdown demonstrates that while factory location and elevation establish baseline performance, operational day-to-day conditions drive massive intra-factory variance.

- **Between vs Within Factory Variance:** One-way ANOVA confirms factory differences are statistically significant (F = 6.08, p = 7.14e-07). Factory means range from 6.07% (Factory_B, Nuwara Eliya) to 7.34% (Factory_H, Low-Country). However, within-factory variation (std 1.1%–1.8%) is substantial, indicating that batch-level controls heavily influence refuse generation.
- **Uva Quality Season Anomaly:** During the South-West Monsoon (June–August), dry foehn (Cachan) winds cross into Uva, dropping mean relative humidity from 73.3% to 57.9% (-15.4% drop). Despite this atmospheric desiccation, refuse tea remains steady (6.29% vs 6.33%), demonstrating regional micro-climatic decoupling.

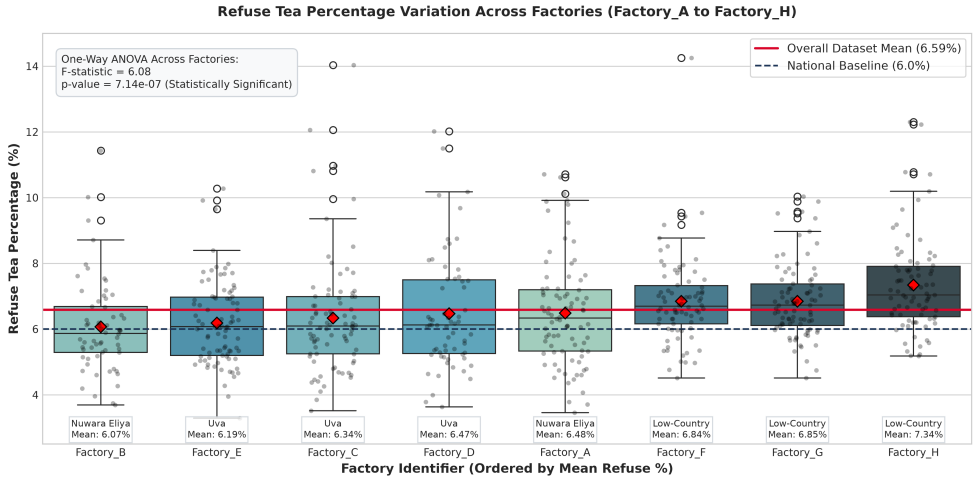


Figure 7: *Factory-level boxplots ordered by mean refuse percentage with ANOVA statistical test.*

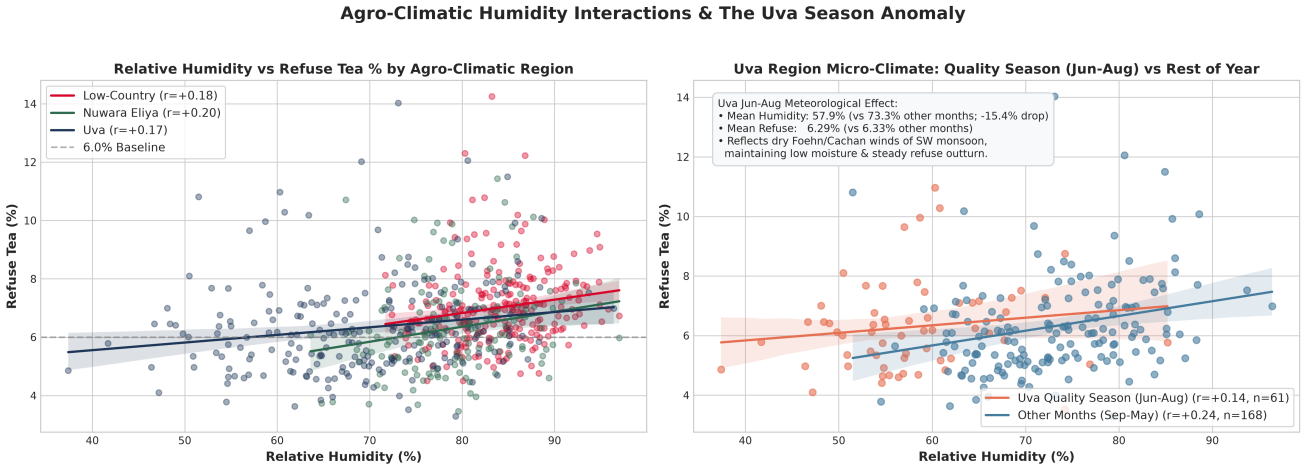


Figure 8: *Agro-climatic humidity interactions highlighting the Uva dry season meteorological shift (Jun–Aug).*

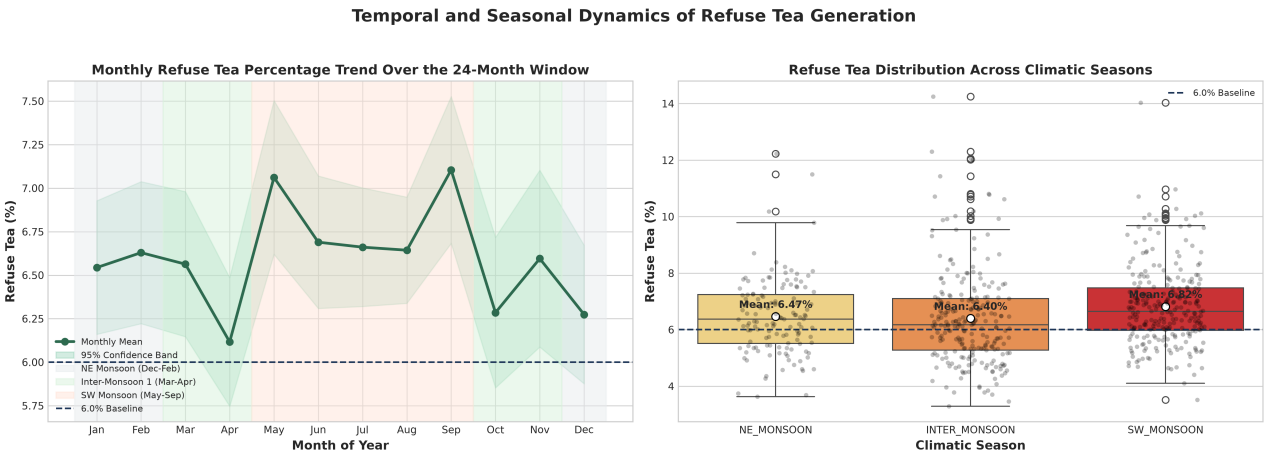


Figure 9: *24-month timeline showing refuse spikes during May and September monsoon rainfall peaks.*

6. Critical Data Leakage Audit

MANDATORY EXCLUSION DIRECTIVE FOR PREPROCESSING:
made_tea_output_kg and refuse_tea_kg are mathematically derived post-hoc scale weights. Feeding either into the training matrix X creates 100% data leakage. **Both columns must be dropped in preprocessing.py.**

In factory accounting and in this synthetic generator, total output mass is the exact sum of made tea and refuse tea:

Total Output (kg) = made_tea_output_kg + refuse_tea_kg

By definition, the target variable is computed algebraically from these end-of-process scale weights:

refuse_tea_pct = [refuse_tea_kg / (made_tea_output_kg + refuse_tea_kg)] × 100

Reconstructing refuse_tea_kg algebraically against recorded mass yields a maximum error of only 0.0156 kg (strictly rounding noise), confirming an exact mathematical identity. Neither weight is known until 24–36 hours after manufacturing completes, making them unviable as operational predictive inputs.

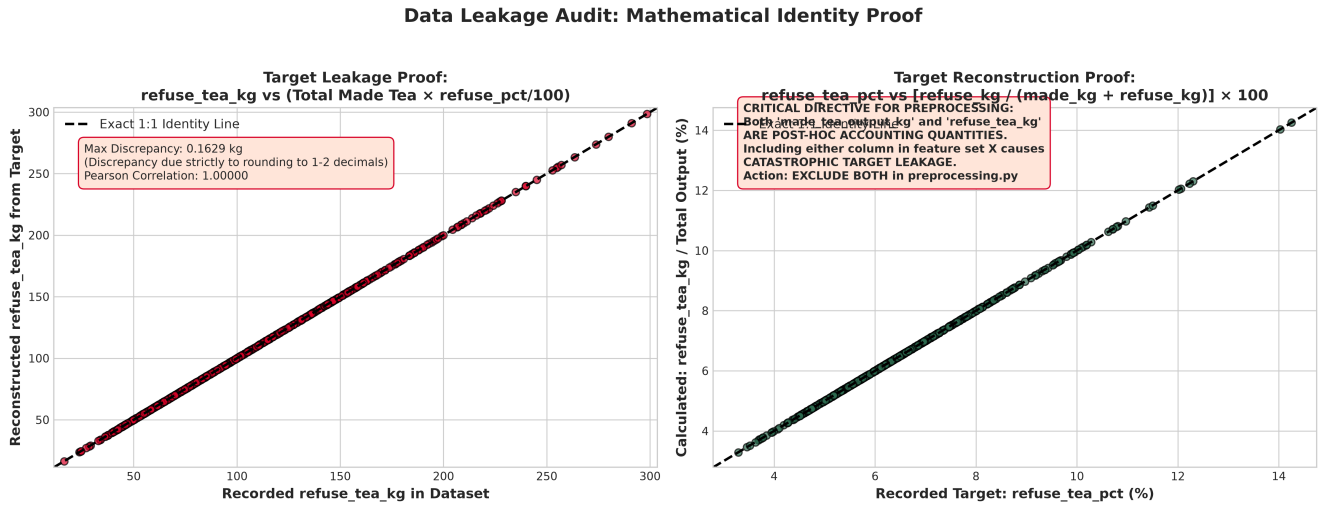


Figure 10: *Algebraic proof of target reconstruction and 100% data leakage verification.*

7. Feature Readiness & Preprocessing Checklist

The feature matrix for the upcoming preprocessing stage is structured as follows:

Feature Name	Role / Status	Signal Tier	Recommended Treatment in preprocessing.py
machine_downtime_hrs	Model Input	Tier 1 (High)	Log1p or RobustScaler to compress 5.5% heavy outlier tail.
withering_duration_hrs	Model Input	Tier 1 (High)	Tree splitting or distance-from-optimum |x - 17.0|.
dryer_temperature_c	Model Input	Tier 1 (High)	Tree splitting or distance-from-optimum |x - 110.0|.
relative_humidity_pct	Model Input	Tier 2 (Med)	StandardScaler; interaction with region/month.
plucking_fine_leaf_pct	Model Input	Tier 2 (Med)	StandardScaler; clean symmetric distribution.
green_leaf_input_kg	Model Input	Tier 2 (Med)	RobustScaler; captures trough volume loading stress.
ambient_temperature_c	Model Input	Tier 2 (Med)	StandardScaler; captures elevation temperature.
region / season / grade	Model Input	Tier 3 (Cat)	One-Hot Encoding (drop_first=True).
elevation_band	**EXCLUDED**	Redundant	Drop; 100% collinear with region (Low=LOW, Uva=MED, Nuwara=HIGH).
made_tea_output_kg	**EXCLUDED**	Leakage	Drop; post-hoc accounting mass causing 100% target leakage.
refuse_tea_kg	**EXCLUDED**	Leakage	Drop; post-hoc accounting mass causing 100% target leakage.
refuse_resale_status	**EXCLUDED**	Post-Hoc	Drop from X; retain for post-prediction revenue impact.
batch_id / batch_date	**EXCLUDED**	Metadata	Drop batch_id; parse batch_date into month and season.

Report Approved: Senior Data Science Team | **Next Milestone:** ml-service/training/preprocessing.py implementation